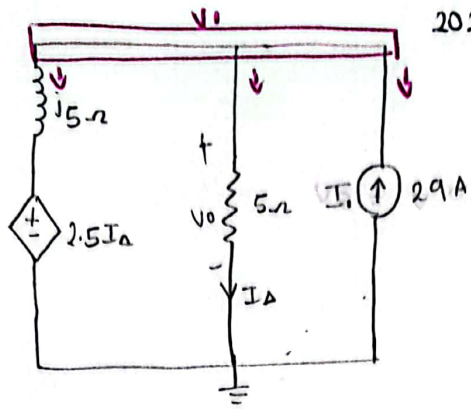


①



$$V_1 = V_0$$

$$I_{\Delta} = \frac{V_1}{5}$$

$$\frac{V_1 - 2.5 I_{\Delta}}{j5} + \frac{V_1}{5} - 29 = 0$$

$$\frac{V_1 - 2.5 \left(\frac{V_1}{5} \right)}{j5} + \frac{V_1}{5} = 29$$

$$\frac{V_1 - 0.5 V_1}{j5} + \frac{V_1}{5} = 29$$

$$\frac{0.5 V_1}{j5} + \frac{V_1}{5} = 29$$

$$V_1 \left(\frac{0.5}{j5} + \frac{1}{5} \right) = 29$$

$$V_1 = \frac{29}{\left(\frac{0.5}{j5} + \frac{1}{5} \right)}$$

$$V_1 = 116.58$$

$$V_0 = 129.69 \angle 26.57^\circ \text{ V}$$

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$$0 = (415) - \frac{3-0V}{10} +$$

$$4 = 415 + \frac{3}{10} - \frac{0V}{10}$$

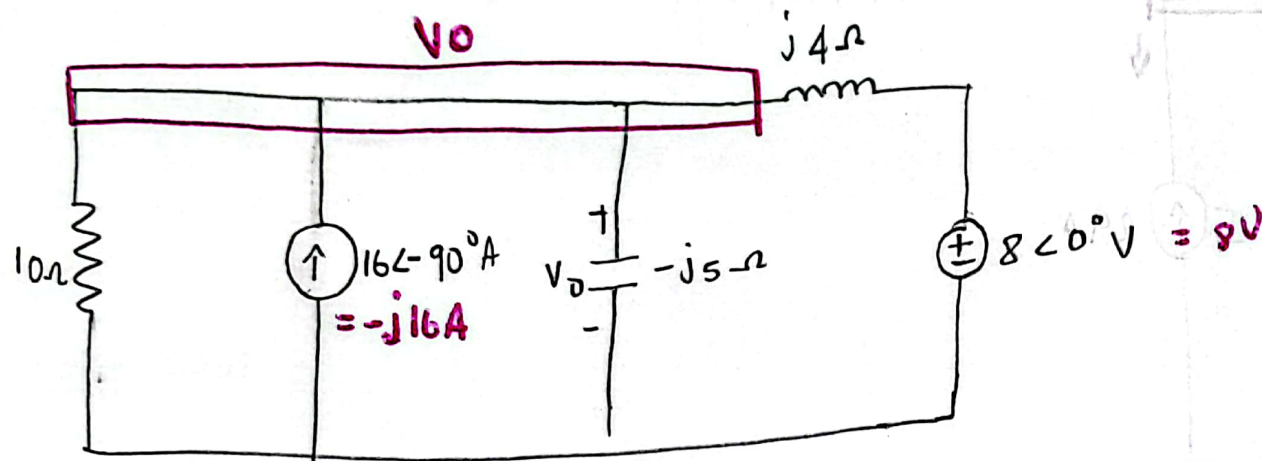
$$415 - \frac{3}{10} = \left(\frac{1}{10} + \frac{1}{20} \right)$$

$$415 - \frac{3}{10}$$

$$\left(\frac{1}{10} + \frac{1}{20} + \frac{1}{20} \right)$$

$$V (\text{ ANI-S })$$

②



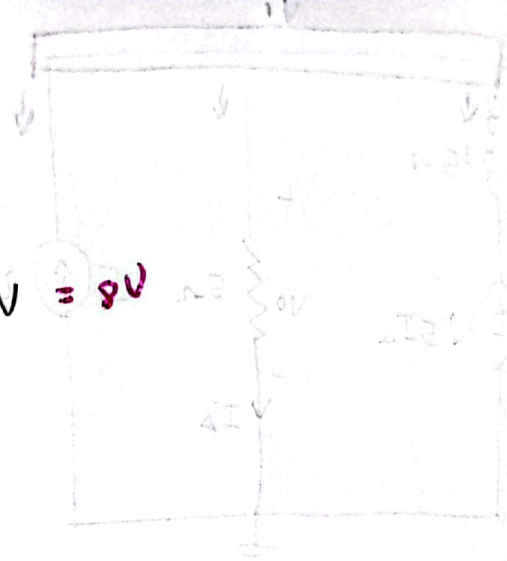
$$\frac{V_0}{10} + \frac{V_0}{-j5} + \frac{V_0 - 8}{j4} - (-j16) = 0$$

$$\frac{V_0}{10} + \frac{V_0}{-j5} + \frac{V_0}{j4} - \frac{8}{j4} + j16 = 0$$

$$V_0 \left(\frac{1}{10} + \frac{1}{-j5} + \frac{1}{j4} \right) = \frac{8}{j4} - j16$$

$$V_0 = \frac{\frac{8}{j4} - j16}{\left(\frac{1}{10} + \frac{1}{-j5} + \frac{1}{j4} \right)}$$

$$V_0 = (-72 - j144) \text{ V}$$



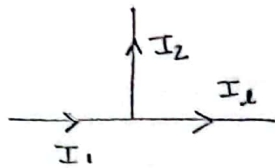
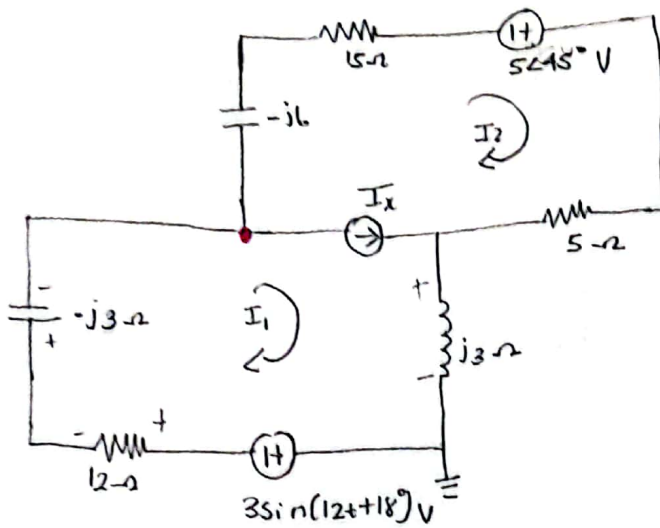
$$0 = \frac{1V}{2} + \frac{\Delta I \cdot 2 - 1V}{2j}$$

$$0 = \frac{1V}{2} + \frac{\left(\frac{1V}{2} \right) \cdot 2 - 1V}{2j}$$

$$0 = \frac{1V}{2} + \frac{1V - 1V}{2j}$$

$$0 = \frac{1V}{2} + \frac{1V - 1V}{2j}$$

③



$$I_{in} = I_{out}$$

$$I_1 = I_2 + I_x$$

$$I_x = I_1 - I_2 \quad \text{--- ①}$$

$$j3I_1 + 3\sin(12t+18^\circ) + 12I_1 - j3I_1 + (-j6I_2) + 15I_2 - 5\angle 45^\circ + 5I_2 = 0$$

$$12I_1 + 20I_2 - j6I_2 = 5\angle 45^\circ - 3\sin(12t+18^\circ)$$

$$12I_1 + I_2(20-j6) = -3 \left[-\frac{5\angle 45^\circ}{3} + \sin(12t+18^\circ) \right]$$

$$\rightarrow \cos(12t-72^\circ)$$

$$\frac{12I_1 + (20-j6)I_2}{\left[\frac{5\angle 45^\circ}{-3} + \cos(12t-72^\circ) \right]} = -3$$

$$\frac{12I_1 + (20-j6)I_2}{\left[\frac{5\angle 45^\circ}{-3} + 1\angle -72^\circ \right]} = -3$$

$$(-1.97 + j4.83)I_1 + (-0.87 + j9.04)I_2 = -3$$

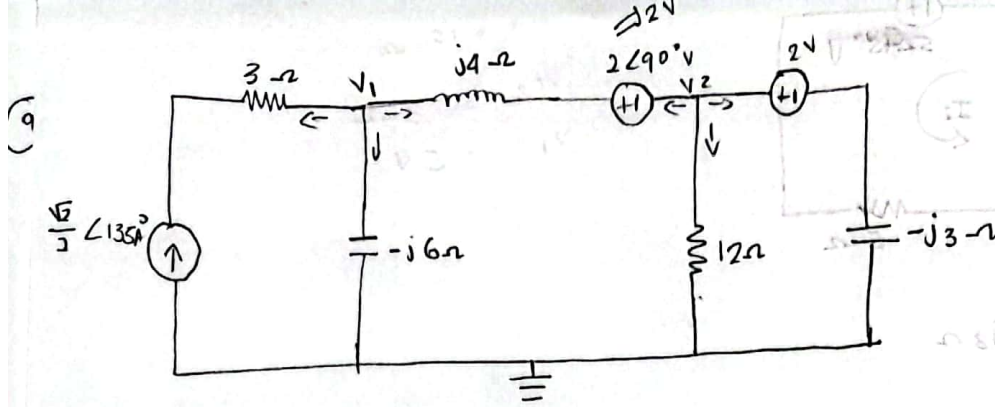
$$\textcircled{a} \alpha = (-1.97 + j4.83)$$

$$\textcircled{b} \beta = (-0.87 + j9.04)$$

$$R_1 = 15 \Omega$$

$$Z_{C2} = 5 \Omega$$

$$V_1 = 5V$$



$$\frac{V_1}{-j6} + \frac{V_1 - j2 - V_2}{j4} - \frac{\sqrt{2}}{2} \angle 135^\circ = 0$$

$$\frac{V_1}{-j6} + \frac{V_1}{j4} - \frac{V_2}{j4} = \frac{\sqrt{2}}{2} \angle 135^\circ + \frac{j2}{j4}$$

$$-j\frac{1}{12}V_1 + j\frac{1}{4}V_2 = j\frac{1}{2}$$

$$-\frac{j}{3}V_1 + jV_2 = j2$$

$$-\frac{1}{3}V_1 + V_2 = 2 \quad \text{--- (1)}$$

$$\frac{V_2 + j2 - V_1}{j4} + \frac{V_2}{12} + \frac{V_2 - 2}{-j3} = 0$$

$$-12i$$

$$-\frac{1}{j4}V_1 + \left(\frac{1}{12} + j\frac{1}{12}\right)V_2 = j\frac{2}{3} - \frac{j2}{j4}$$

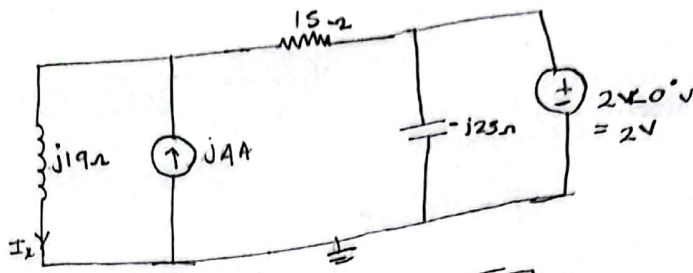
$$j\frac{1}{4}V_1 + \left(\frac{1}{12} + j\frac{1}{12}\right)V_2 = -\frac{1}{2} + j\frac{2}{3}$$

$$3V_1 + (1 - j)V_2 = 8 + 6i \quad \text{--- (2)}$$

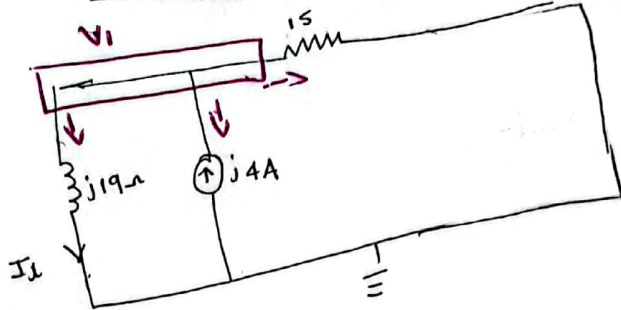
(5) $a = -0.33$ * $\rightarrow$ THIS WAS WRONG & I DK HOW WHAT THE SOLUTION IS.
 (6) $b = 1$
 (7) $c = 3$
 (8) $d = (1-j)$

$$11\angle 0^\circ = 11V$$

9



$$\begin{aligned} L &= 19 \mu H \\ R_1 &= 15 \Omega \\ C &= 25 \mu F \\ I_1 &= 4 A \end{aligned}$$



$$I_x = \frac{V_1}{j19}$$

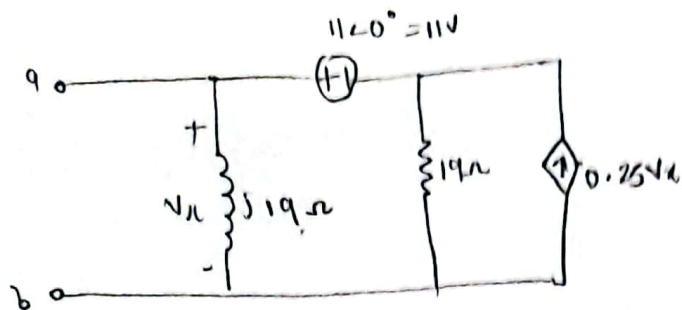
$$\frac{V_1}{j19} - j4 + \frac{V_1}{15} = 0$$

$$V_1 \left(\frac{1}{j19} + \frac{1}{15} \right) = j4$$

$$V_1 = \frac{j4}{\left(\frac{1}{j19} + \frac{1}{15} \right)} V$$

$$I_x = \frac{\frac{j4}{\left(\frac{1}{j19} + \frac{1}{15} \right)}}{j19}$$

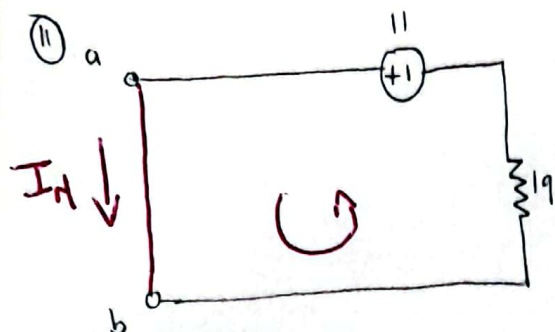
$$I_x = 1.95 + j1.54 A$$



$$P_g = 19 \text{ W}$$

$$L = 19 \text{ mH}$$

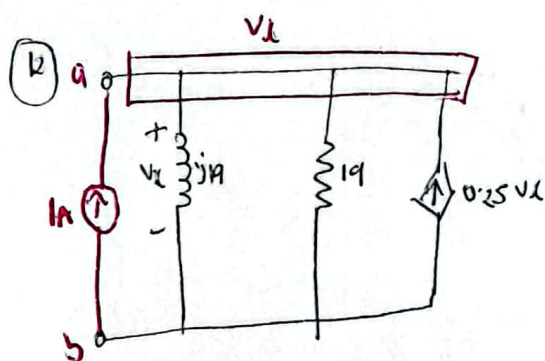
$$V_s = 15 \text{ V}$$



$$19I_N - 11 = 0$$

$$I_N = \frac{11}{19}$$

$$I_N = 0.579 \text{ A}$$



$$\frac{V_x}{j19} + \frac{V_x}{19} - 0.25V_x - 1 = 0$$

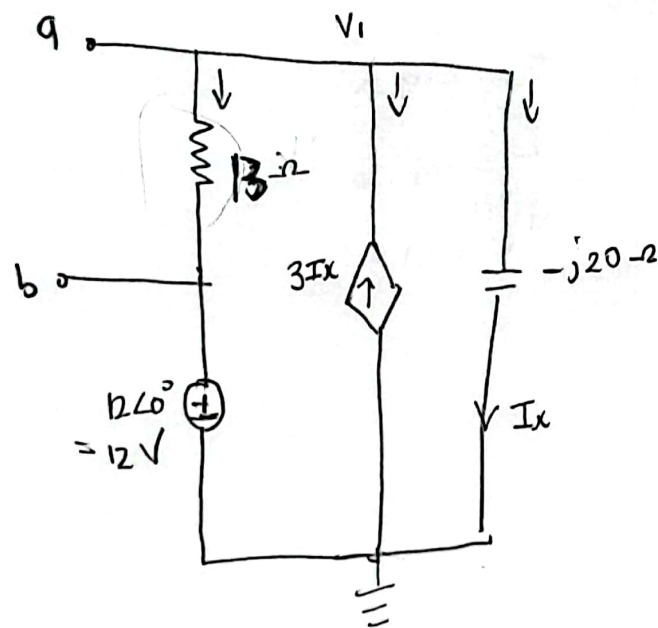
$$V_x \left(\frac{1}{j19} + \frac{1}{19} - 0.25 \right) = 1$$

$$V_x = \frac{1}{\frac{1}{j19} + \frac{1}{19} - 0.25} \text{ V}$$

$$Z_N = \frac{1}{\frac{1}{j19} + \frac{1}{19} - 0.25} \text{ V}$$

1 A

$$Z_N = -4.73 + j12.6 \Omega$$



$$V_s = 12V$$

$$R = 13\Omega$$

$$(13) \quad I_x = \frac{V_1}{-j20}$$

$$\frac{V_1 - 12}{13} + 3\left(\frac{V_1}{-j20}\right) + \frac{V_1}{-j20} = 0$$

$$V_1 \left(\frac{1}{13} - \frac{3}{-j20} + \frac{1}{-j20} \right) = \frac{12}{13}$$

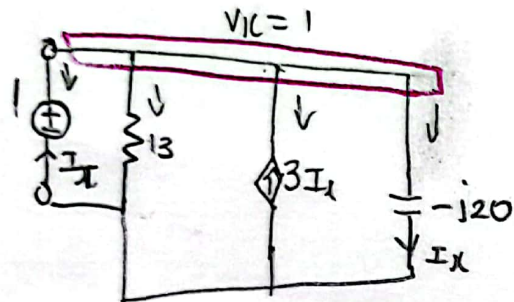
$$V_1 = \frac{12}{13} \div \left(\frac{1}{13} - \frac{3}{-j20} + \frac{1}{-j20} \right)$$

$$V_{TH} = \frac{12}{13} \div \left(\frac{1}{13} - \frac{3}{-j20} + \frac{1}{-j20} \right) - 12$$

$$V_{TH} = (-7.54 + j5.8)V$$

$$V_{TH} = (-7.54 + j5.8)V$$

(14)



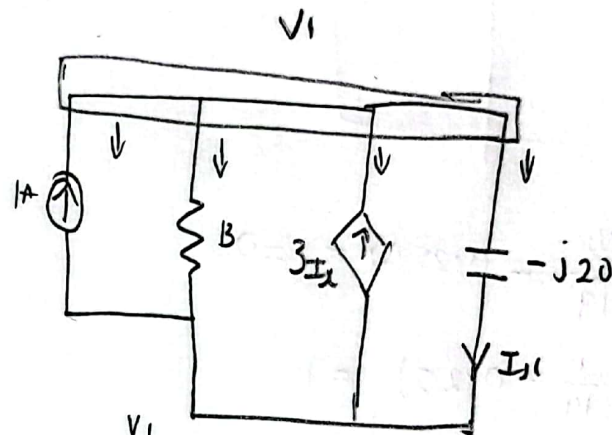
$$I_x = \frac{V_x}{-j20}$$

$$-I_T + \frac{V_1}{13} - 3I_x + \frac{V_1}{-j20} = 0$$

$$\frac{1}{13} - 3\left(\frac{1}{-j20}\right) + \frac{1}{-j20} = \frac{V_T}{V_1}$$

$$Z_{TH} = \frac{1V}{\frac{1}{13} - 3\left(\frac{1}{-j20}\right) + \frac{1}{-j20}}$$

$$Z_{TH} = 4.83 + j6.28 \Omega$$



$$I_x = \frac{V_1}{-j20}$$

$$-1 + \frac{V_1}{13} - 3\left(\frac{V_1}{-j20}\right) + \frac{V_1}{-j20} = 0$$

$$V_1 \left(\frac{1}{13} - \frac{3}{-j20} + \frac{1}{-j20} \right) = 1$$

$$V_1 = \frac{1}{\frac{1}{13} - \frac{3}{-j20} + \frac{1}{-j20}}$$

$$Z_{TH} = \frac{1}{\frac{1}{13} - \frac{3}{-j20} + \frac{1}{-j20}} = (4.83 + j6.28) \Omega$$